Titanic Survival Prediction

Problem: predict the binary Survived outcome (0 = did not survive, 1 = survived) from passenger-manifest information.

Dataset: 891 passengers and 12 source columns. Verified columns: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked.

Missing values in this CSV: Age=177, Cabin=687, Embarked=2. Cabin was excluded due to its high missingness; no passenger rows were removed.

Methodology: stratified 80/20 train-test split with random_state=42. Median numeric and most-frequent categorical imputation, one-hot encoding, and scaling where appropriate are fitted only within training pipelines.

Feature engineering: family size, alone indicator, grouped name title, and log1p fare. IQR flags were inspected but passengers were not removed; log1p was used to reduce fare skew.

Actual holdout test results

	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.844	0.825	0.754	0.788
Random Forest	0.821	0.785	0.739	0.761
Gaussian Naive Bayes	0.81	0.74	0.783	0.761

Algorithms and Model Selection

Logistic Regression: linear probabilistic classifier; numeric features scaled.

Random Forest: ensemble of 300 decision trees; numeric features not scaled.

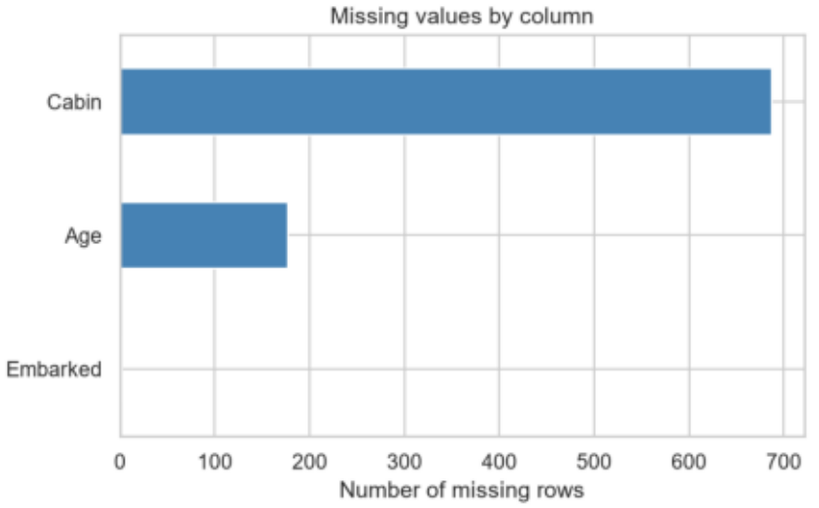
Gaussian Naive Bayes: probabilistic classifier with a conditional-independence assumption; numeric features scaled and categorical encoding densified.

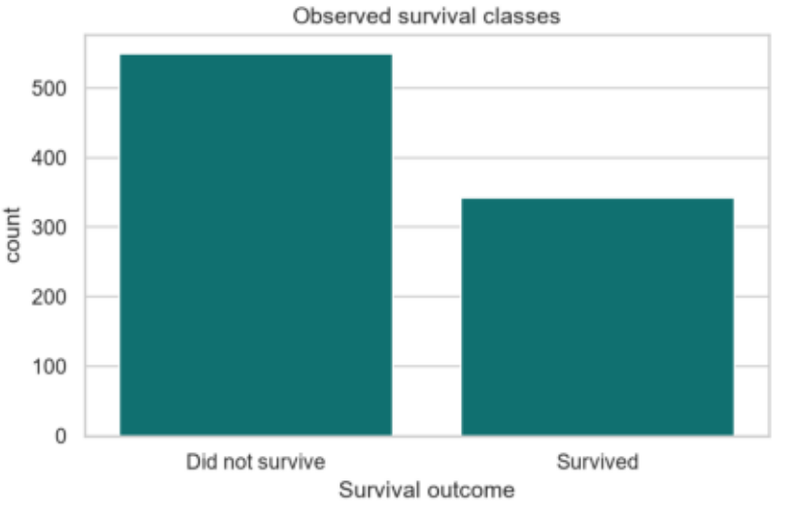
Selection rule: highest holdout F1-score, then recall, then accuracy. The selected model is Logistic Regression, with accuracy 0.844, precision 0.825, recall 0.754, and F1-score 0.788.

Limitations: this is a small historical dataset and a single holdout split. The results are not external validation or a guarantee of future performance. High Cabin missingness and simplified feature engineering may limit prediction.

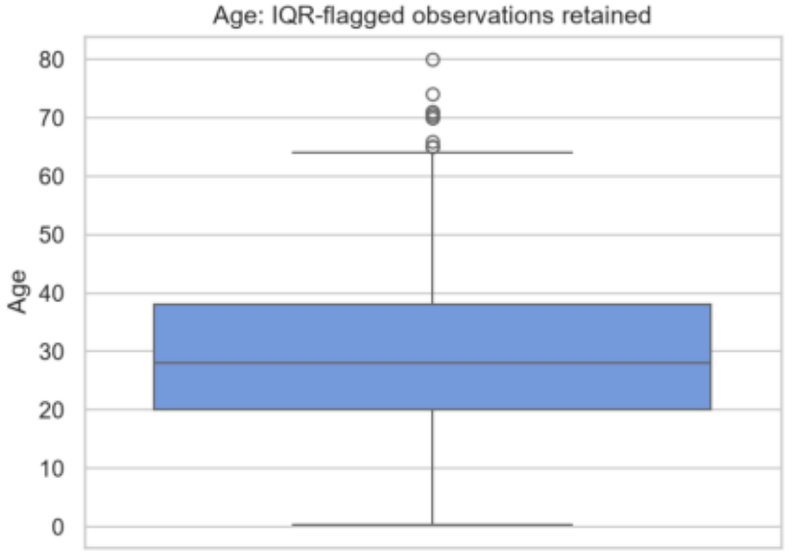
Conclusion: the selected model is the strongest under the stated metric rule on this test split. More validation and careful assessment of generalization would be needed before real-world use.

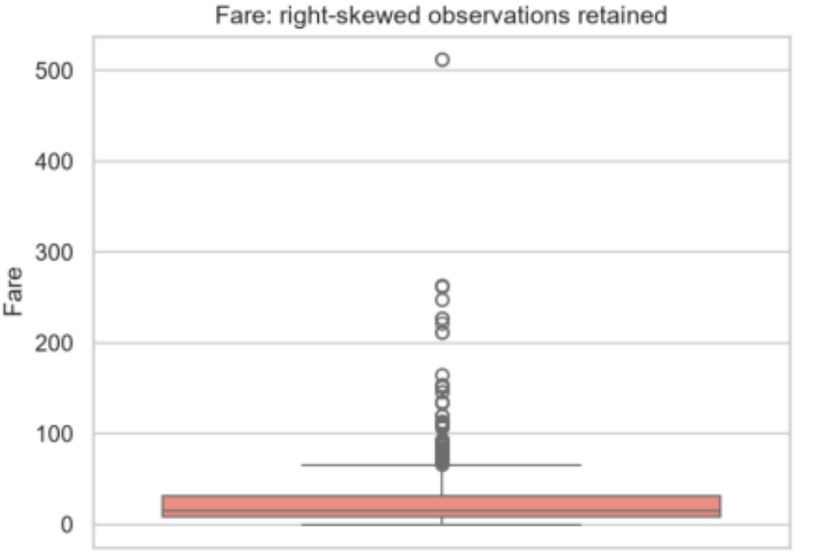
Dataset Overview





Outlier Analysis





Confusion Matrices

